



Scheelite

△ Specimen of scheelite in a rock groundmass

Crystals of scheelite can be opaque or transparent. The latter are sometimes cut as gemstones for collectors, and these exhibit almost as much dispersion (fire) as diamond. For this reason, synthetic colourless scheelite is sometimes used as a diamond simulant, although it is too soft to wear well. Synthetic scheelite is also coloured by trace elements to simulate other gemstones. Opaque crystals can grow very large – some weighing up to 7kg (15lb) come from Arizona, USA. Most scheelite crystals fluoresce under ultraviolet light.

Specification

Chemical name	Calcium tungstate	Formula	CaWO_4
Colours	Yellow, white, pale green, orange	Structure	Tetragonal
Hardness	4.5–5	SG	5.9–6.3
		RI	1.92–1.94
		Lustre	Vitreous to greasy
		Streak	White
		Locations	Austria, Italy, Brazil, Rwanda, USA, UK, China



Scheelite on muscovite | Rough | In this specimen originating from China, a large crystal is seated on a groundmass of muscovite, a common variety of mica.

Scheelite crystal



Bipyramidal crystal

Crystals in matrix | Rough | This specimen consists of scheelite crystals resting on a groundmass of magnetite. The crystals are bipyramidal in shape.



Good transparency

Gem crystal | Rough | This finely formed, gemmy scheelite crystal originates from a major scheelite deposit location in China. Stones such as this can be cut into gems.

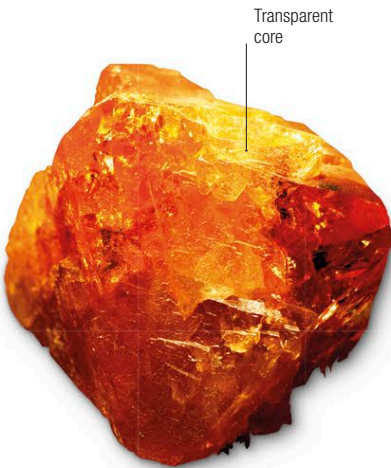
Tungsten

Turning up the heat

Scheelite is a major source of tungsten, which has the highest melting point of all elements and is a vital part of modern industry. Electric light filaments are made from pure tungsten, while tungsten carbide is used in drill bits, dies, and tools for shearing metal; cobalt-chromium-tungsten alloys are used in the surfaces of wear-resistant valves, bearings, propeller shafts, and cutting tools. Tungsten steel is used in high-temperature hardware such as rocket nozzles.



Rocket nozzle High-temperature alloys such as this rocket nozzle depend on tungsten derived from scheelite.



Transparent core

Orange crystal | Colour variety | This large, finely transparent crystal is notable for its vivid orange colouring. A stone such as this would fluoresce under ultraviolet light.



Crown facets

Brilliant cut | Cut | The cut of this fine, brilliant-cut, yellow-brown scheelite gem enhances the high reflectivity associated with this type of mineral.